# From Unequal Beginnings to Social Starting Capital

## A Normative and Budget-Closed Framework for Common Inheritance, Child Development, and Protected Capital

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**Abstract**

People do not choose the families, assets, legal systems, public institutions, or historical circumstances into which they are born. Yet those conditions shape their capabilities long before individual effort can plausibly explain the resulting differences. This paper argues that part of a society's productive inheritance—the value attributable to nature, accumulated knowledge, public institutions, social cooperation, and historically created scarcity rents—may legitimately finance a material starting stake for every child. It then develops the Social Starting Capital Framework (SSCF), a policy architecture with three distinct functions: a universal child-development floor, supplementary household support, and an independently owned birthright capital account.

The formal model is designed to be budget-closed. Every newborn receives an equal, strictly positive statutory base grant. Family size affects supplementary household support, not the child's independent capital right. All discretionary bonuses and progressive top-ups are normalized to explicit pools, so realized expenditure cannot exceed the authorized budget when births or household composition differ from forecasts. The model distinguishes maximum spending capacity from actual spending, specifies the split between current revenue and fund withdrawals, includes withdrawals in child-account dynamics, and states an exact condition for preserving the social wealth fund in real per-capita terms. A fiscal backstop makes the claimed minimum a genuine guarantee rather than a residual aspiration. A calibration and demographic stress test illustrate the system.

The proposal is neither a theory of equal outcomes nor a substitute for public services, private saving, or claims of historical rectification. It is a pre- distributional institution: each generation inherits a protected material foothold in a social world it did not create but must be able to enter as an agent.

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# Part I
# The Normative Case

## 1 Introduction

The conditions of birth are unchosen, but they are not inconsequential. A child may be born into secure housing, educated networks, inherited assets, political stability, and ready access to credit. Another may be born into debt, displacement, dangerous housing, weak public services, or legal insecurity. Neither child authored that initial position. By the time conventional accounts of merit begin measuring effort, the two children may already differ in health, vocabulary, confidence, time horizon, and the range of choices they can safely make.

This fact does not imply that every later inequality is unjust. Differences can arise from effort, preference, innovation, luck, care obligations, and voluntary exchange. The narrower claim defended here is that a political order cannot treat radically unequal starting power as morally irrelevant and then describe the resulting competition as equal opportunity. Formal permission is not the same as effective freedom. A right to study, move, refuse an abusive employer, or start an enterprise is thin when a person lacks the health, time, security, and assets required to exercise it.

The empirical background is severe. The *World Inequality Report 2022* estimates that the richest tenth of the global population receives roughly 52 percent of global income and owns roughly 76 percent of global wealth, while the poorest half receives 8.5 percent of income and owns about 2 percent of wealth [21]. Land—a finite and politically constituted asset—is also highly concentrated; in the most unequal regions, the top tenth of landowners controls as much as three quarters of agricultural land [3]. Measures of intergenerational mobility differ by country and by whether one studies income, education, absolute mobility, or relative mobility, but family background remains a powerful predictor of later outcomes [19].

The policy response developed in this paper begins from a concept of *common inheritance*. Each generation receives natural resources, scientific knowledge, language, infrastructure, legal forms, monetary institutions, and productive organizations that it did not build from nothing. Those inheritances do not make all wealth collective. They do, however, weaken the claim that the entire social product is morally exhausted by present private title.

The Social Starting Capital Framework (SSCF) turns that moral observation into an institution with three separate layers:

1. a universal child-development floor, delivered primarily through services or guaranteed provision;

2. supplementary household support, recognizing that care occurs in families and has both fixed and variable costs; and

3. a protected capital account owned by the child, containing an equal statutory base grant and, where authorized, budget-normalized additions.

Separating these layers is essential. Services cannot provide the bargaining power of an owned asset. A capital account cannot replace nutrition, healthcare, safe housing, or schools. Household assistance is necessary, but a payment controlled entirely by parents is not the same institution as capital legally owned by the child.

The remainder of Part I develops the moral argument and its limits. Part II provides a complete accounting model. Part III gives a calibration, stress test, governance rules, and limitations.

## 2 Unequal beginnings and effective freedom

The metaphor of the birth lottery is useful because it identifies a boundary of responsibility. No one chooses their parents, citizenship, childhood neighborhood, inherited health, first language, or initial balance sheet. Rawls treats natural and social starting positions as morally arbitrary for the purpose of choosing principles for the basic structure [15]. Sen shifts attention from formal holdings to capabilities: what a person is substantively able to be and do [17]. Van Parijs similarly distinguishes formal freedom from the resources required for real freedom [20].

Capital matters within this framework for reasons different from income. Income meets recurrent needs. Public services create capabilities and insure risks. Capital adds a stock of option value. It can finance relocation, education, tools, a housing deposit, or a period of search. It can make refusal possible. An individual with a modest asset buffer can reject a predatory loan or dangerous job in a way that an otherwise equal individual living at the edge of subsistence cannot.

This paper therefore uses *starting capital* in the financial and institutional sense. It should not be confused with the sociological concept of *social capital*, which normally refers to relationships, networks, trust, and norms. Networks are part of unequal beginnings, but the SSCF is a proposal for social starting *capital*: a publicly constituted material stake.

## 3 Property, history, and common inheritance

Modern holdings did not emerge on an empty field. They are the result of labor and exchange, but also of inheritance, enclosure, conquest, slavery, colonial extraction, state-building, technological accumulation, and legal decisions about which claims the state will enforce. Recognizing this history does not establish that every present title is invalid. It establishes that present title cannot be the only premise in a theory of distribution.

Locke's labor theory of appropriation remains important because effort and personal projects deserve protection. Yet Lockean appropriation is constrained by the proviso that enough and as good remain for others [6]. Once land and resources are scarce and legally enclosed, latecomers encounter neither open nature nor a blank slate. Rousseau emphasizes that civil inequality is institutional, not merely a reflection of natural difference [16]. Paine argues that private cultivation can be recognized while those excluded from the earth's original common condition receive compensation [12]. George makes a related distinction between productive effort and land rent generated by scarcity and surrounding social development [4].

Marx's account of primitive accumulation and Polanyi's account of land, labor, and money as politically constructed commodities place market exchange within a prior history of institutional change [7, 14]. Their analyses need not be accepted in full to recognize that markets depend on law, public authority, infrastructure, money, and accumulated knowledge.

Nozick's entitlement theory offers the strongest warning against treating persons and their holdings as instruments of a preferred pattern. Just holdings acquired and transferred legitimately deserve protection [9]. But the historical principle also makes acquisition and rectification unavoidable questions. A society with layered dispossession, inherited legal privilege, and uncompensated externalities cannot simply assume the clean history that gives the entitlement argument its greatest force.

The conclusion is limited but significant: a defined portion of economic flows can be treated as socially claimable without treating all property or production as collective. The relevant base is not "everything the state can tax." It is a transparent set of revenues connected to common assets, inherited transfers, publicly created value, external costs, and collectively owned capital.

This conclusion is strengthened by two further observations. First, wealth can compound across generations and become increasingly important to stratification when returns on owned assets outpace the growth of ordinary incomes [13]. Second, recognizing a common claim does not require a choice between atomized private ownership and unrestricted central control. Ostrom's work demonstrates that shared resources can be governed through durable, rule-bound institutions under the right conditions [11]. The design question is therefore not simply public versus private, but which rights, duties, boundaries, and monitoring arrangements fit the resource.

# 4 Three arguments that must remain distinct

Several arguments for redistribution are often blended together. They should be separated because they imply different allocation rules.

## 4.1 Common inheritance

If every member of a generation is a co-beneficiary of natural, institutional, and knowledge inheritances, the common-inheritance argument supports a universal claim. The equal base grant in this paper rests primarily on that argument.

## 4.2 Capability and priority

Equal nominal grants do not equalize substantive freedom. Disability, regional prices, discrimination, care burdens, and household poverty affect what the same amount can do. Capability and prioritarian arguments therefore support progressive or needs-based top-ups. Those additions should be explicit and separately budgeted rather than hidden inside the universal claim.

## 4.3 Rectification

Historical injustice can generate claims for groups or persons who suffered or inherit the consequences of a specific wrong. Such claims may be larger, differently funded, or differently governed than a universal starting stake. A universal SSCF must not be presented as extinguishing claims for reparations, restitution, indigenous sovereignty, or other forms of rectification.

This separation improves both philosophy and policy. Universality, progressivity, and rectification can coexist, but one does not silently substitute for another.

# 5 Intellectual and institutional lineage

The proposal belongs to a long family of arguments. Paine proposed a national fund that would pay a capital sum at adulthood [12]. More recent proposals include stakeholder grants, minimum inheritance, and asset-building policies [1, 2, 18]. Hamilton and Darity's baby-bonds work shows how progressively funded child accounts can address entrenched wealth gaps [5].

Public wealth funds provide a second lineage. Norway's Government Pension Fund turns petroleum revenue into long-term public saving and explicitly links its management to intergenerational considerations [8]. These examples do not by themselves justify the SSCF, nor are they institutionally identical to it. They establish that public ownership, professional asset management, and intergenerational distribution are practical institutional categories rather than conceptual impossibilities.

The SSCF's contribution is architectural. It integrates present capabilities, family care, individually owned capital, explicit fiscal priority, and a fund rule in one budget-closed model. Its novelty claim is therefore about the combination and the formal separation of functions, not the invention of child accounts or public funds.

## 6 Normative principles

The framework is governed by seven principles.

**P1. Common inheritance.**
A defined part of social wealth is attributable to nature, inherited institutions, public action, and cumulative cooperation rather than current private effort alone.

**P2. Independent child standing.**
The child is a rights-bearing unit. Every eligible newborn receives an equal, strictly positive base capital grant that is not reduced by sibling count, parental income, or module completion.

**P3. Capability before discretion.**
Essential childhood development has fiscal priority over optional distributions.

**P4. Family agency without substitution.**
Families receive room and support to care, but household transfers neither replace the child-development floor nor absorb the child's capital right.

**P5. Exact budget closure.**
Every variable expenditure is assigned to a defined pool. Forecast errors may change bonuses, but cannot create unauthorized expenditure.

**P6. Intergenerational reciprocity.**
Fund policy is evaluated in real per-capita terms, not merely by nominal non-decline.

**P7. Non-forfeiture and proportionate protection.**
Education may affect the form and timing of early access, but never ownership. Restrictions expire and are subject to due process.

# Part II
# The Formal Framework

## 7 Timing, populations, and notation

Time is discrete and indexed by fiscal years $t = 0, 1, 2, \ldots$. All transfers for birth cohort $t$ are finalized after the eligible births in that cohort have been registered. A provisional account may be opened at birth, but the final annual bonus is credited only after the realized cohort is known. This convention eliminates the use of future or unobserved population values in current grants.

Let:

| Symbol | Meaning |
|---|---|

| | |
|---|---|
| $N_t$ | reference population at the start of year $t$ |
| $\mathcal{F}_t$ | set of eligible households during year $t$ |
| $K_{f,t}$ | number of eligible dependent children in household $f$ |
| $\mathcal{C}_t$ | set of all eligible dependent children |
| $\mathcal{B}_t$ | finalized set of eligible newborns in cohort $t$ |
| $B_t$ | realized births, $B_t = \lvert\mathcal{B}_t\rvert$ |
| $Y_t$ | net, non-overlapping socially claimable inflow |
| $W_t$ | market value of the social wealth fund at the start of year $t$ |
| $r_t$ | net nominal return on the fund during year $t$ |
| $P_t$ | general price index, used only for real-value comparisons |
| $U_t$ | expenditure on the universal child-development floor |
| $G_t^{\min}$ | aggregate statutory newborn base grants |
| $H_t$ | actual supplementary household allocation |
| $G_t^{+}$ | cohort bonus above the statutory base |
| $T_t$ | progressive top-up pool for existing child accounts |
| $V_t$ | optional adult or transition distribution |
| $D_t$ | actual total program distribution |
| $D_t^Y, D_t^W$ | actual distribution funded from current resources and the fund |
| $Z_t$ | transparent general-revenue backstop for mandatory child claims |
| $A_{c,s}$ | total account assets owned by child $c$ at date $s$ |

We assume $N_t > 0$, $P_t > 0$, $W_t \geq 0$, and $B_t \geq 0$. Every formula that divides by $B_t$ includes a separate zero-birth convention.

# 8 The revenue base and social wealth fund

**Definition 8.1** (Socially claimable inflow)**.** *The net inflow dedicated to the framework is*

$$Y_t = Y_t^{\text{res}} + Y_t^{\text{land}} + Y_t^{\text{inh}} + Y_t^{\text{pub}} + Y_t^{\text{eco}} + Y_t^{\text{other}}, \tag{1}$$

*where the terms denote, respectively, net resource rents, land-value revenue, dedicated inheritance and gift taxation, distributable profit from public assets, ecological charges, and other legislatively designated revenue.*

The components must be measured net of collection costs, rebates, and prior legal claims. They must also be non-overlapping. In particular, $Y_t^{\text{pub}}$ excludes returns already counted in $r_t W_t$, and ecological charges exclude resource royalties already counted as resource rents. This restriction prevents the same economic flow from financing the model twice.

Let $\phi_t \in [0, 1]$ be the minimum saving share of $Y_t$. Let $\widetilde{W}_t$ be a prudential reference value known at the start of the year, such as the lesser of current fund value and a trailing multi-year average. With draw-rate parameter $\gamma_t \in [0, 1)$, the maximum authorized fund draw is

$$\overline{d}_t = \min\{\gamma_t \widetilde{W}_t, W_t\}. \tag{2}$$

The use of a predetermined reference value avoids making current entitlements depend on an end-of-year return that is not yet known.

Before the statutory backstop is applied, maximum current-year spending capacity is

$$C_t^0 = (1 - \phi_t) Y_t + \overline{d}_t. \tag{3}$$

This is capacity, not actual expenditure and not an instruction to take the full draw.

## 9 Mandatory child claims and the fiscal backstop

Let $u_{c,t} \geq 0$ be the cost of the guaranteed development package for child $c$. It may reflect disability, regional cost, or age while preserving a common service standard. Then

$$U_t = \sum_{c \in \mathcal{C}_t} u_{c,t}. \tag{4}$$

The package can include nutrition, preventive healthcare, schooling, safe transport, digital access, and housing-related support. It should be measured as the incremental cost attributable to the SSCF or explicitly identify existing expenditure reclassified under it; otherwise the fiscal cost would be overstated or double-counted.

Let $\underline{g}_t > 0$ be the indexed statutory base grant for each eligible newborn. The aggregate base obligation is

$$G_t^{\min} = B_t \underline{g}_t. \tag{5}$$

The mandatory claim is

$$M_t = U_t + G_t^{\min}. \tag{6}$$

A right cannot honestly be called guaranteed if it disappears whenever residual funds are exhausted. The framework therefore makes any shortfall visible:

$$Z_t = \left[ M_t - C_t^0 \right]_+ . \tag{7}$$

Here $Z_t$ is a transfer from ordinary public revenue authorized by the statute creating the SSCF. It may be financed through a standing appropriation or an automatic surcharge. It may not be financed by breaching the fund draw cap. Total authorized capacity is therefore

$$C_t = C_t^0 + Z_t, \tag{8}$$

and $M_t \leq C_t$ by construction.

**Remark 9.1.** *The backstop is not free money. It states who bears the risk when dedicated revenues fall short. Without it, either the minimum is not a right or the model contains an unreported deficit. A legislature may set a modest $\underline{g}_t$ to make activation rare, but it cannot claim an unconditional positive grant while assigning all revenue risk to the child.*

## 10 Supplementary household allocation

Households face fixed costs and economies of scale. Let the desired supplemental amount for household $f$ be

$$\widehat{H}_{f,t} = h_t + s_t \omega(K_{f,t}), \qquad h_t, s_t \geq 0, \tag{9}$$

where

$$\omega(K) = \sum_{j=1}^{K} w_j, \qquad w_j = \begin{cases} 1, & j \leq K^*, \\ \lambda^{j-K^*}, & j > K^*, \end{cases} \tag{10}$$

with $K^* \in \{1, 2, \ldots\}$ and $0 < \lambda < 1$. The taper applies only to this supplementary household layer. It never changes $\underline{g}_t$.

Let

$$\widehat{H}_t = \sum_{f \in \mathcal{F}_t} \widehat{H}_{f,t} \tag{11}$$

be desired aggregate household support. A share $\beta_H \in [0,1]$ of capacity remaining after mandatory claims is reserved for this layer:

$$\overline{H}_t = \beta_H (C_t - M_t). \tag{12}$$

Define

$$\xi_t = \begin{cases} \min\{1, \overline{H}_t / \widehat{H}_t\}, & \widehat{H}_t > 0, \\ 0, & \widehat{H}_t = 0, \end{cases} \tag{13}$$

and set

$$H_{f,t} = \xi_t \widehat{H}_{f,t}, \qquad H_t = \sum_{f \in \mathcal{F}_t} H_{f,t}. \tag{14}$$

Thus household support is both individually tapered and aggregately budgeted.

**Proposition 10.1** (Household bounds)**.** *For every household $f$,*

$$0 \le H_{f,t} \le h_t + s_t \left( K^* + \frac{\lambda}{1-\lambda} \right), \tag{15}$$

*and aggregate expenditure satisfies $0 \le H_t \le \overline{H}_t$.*

*Proof.* For $K \le K^*$, $\omega(K) = K \le K^*$. For $K > K^*$,

$$\omega(K) = K^* + \sum_{m=1}^{K-K^*} \lambda^m \le K^* + \frac{\lambda}{1-\lambda}.$$

Since $0 \le \xi_t \le 1$, the individual bound follows. The definition of $\xi_t$ gives $H_t = \min\{\widehat{H}_t, \overline{H}_t\}$ when $\widehat{H}_t > 0$, proving the aggregate bound. □

Because the base term $h_t$ could create incentives to split one economic household into multiple administrative units, eligibility must rely on a legally defined care and residence unit, with due-process protections for genuine family separation.

## 11 Residual pools and exact allocation

After mandatory claims and actual household support, the remaining capacity is

$$Q_t = C_t - M_t - H_t \ge 0. \tag{16}$$

Choose policy shares $\theta_G, \theta_T, \theta_V \ge 0$ satisfying

$$\theta_G + \theta_T + \theta_V \le 1. \tag{17}$$

The authorized pools are

$$\overline{G}_t^+ = \theta_G Q_t, \qquad \overline{T}_t = \theta_T Q_t, \qquad \overline{V}_t = \theta_V Q_t. \tag{18}$$

### 11.1 Equal newborn bonus

The cohort bonus is

$$b_t^+ = \begin{cases} \overline{G}_t^+ / B_t, & B_t > 0, \\ 0, & B_t = 0. \end{cases} \tag{19}$$

Every newborn $c \in \mathcal{B}_t$ receives

$$g_{c,t}^{(0)} = \underline{g}_t + b_t^+. \tag{20}$$

Consequently,

$$\sum_{c \in \mathcal{B}_t} g_{c,t}^{(0)} = G_t^{\min} + \overline{G}_t^+ \tag{21}$$

when $B_t > 0$. The realized cohort, rather than smoothed or forecast births, is the final denominator. Forecasts may be used for budgeting, but never to calculate an entitlement that can overspend its pool.

### 11.2 Progressive account top-ups

Let $y_{c,t} \geq 0$ be an administratively defined measure of the resources available to child $c$, preferably averaged across several years and adjusted for household size. Let $y_t^* > 0$ be a reference threshold. Define the bounded shortfall score

$$z_{c,t} = \mathrm{clamp}_{[0,1]}\left( \frac{y_t^* - y_{c,t}}{y_t^*} \right). \tag{22}$$

Additional bounded indicators may represent disability, care burden, displacement, or regional cost. For clarity, combine them into $d_{c,t} \in [0,1]$ and define

$$a_{c,t} = 1 + \kappa_y z_{c,t} + \kappa_d d_{c,t}, \qquad \kappa_y, \kappa_d \geq 0. \tag{23}$$

If $\mathcal{C}_t$ is nonempty, allocate

$$\tau_{c,t} = \overline{T}_t \frac{a_{c,t}}{\sum_{j \in \mathcal{C}_t} a_{j,t}}. \tag{24}$$

If $\mathcal{C}_t$ is empty, set every $\tau_{c,t} = 0$ and $T_t = 0$. Otherwise set $T_t = \sum_c \tau_{c,t}$.

**Proposition 11.1** (Exact pool exhaustion)**.** *If $\mathcal{C}_t \neq \varnothing$, then $\tau_{c,t} > 0$ for every eligible child and*

$$\sum_{c \in \mathcal{C}_t} \tau_{c,t} = \overline{T}_t. \tag{25}$$

*No income report or demographic realization can cause top-ups to exceed their pool.*

*Proof.* Since $a_{c,t} \geq 1$, every denominator and every individual weight is positive. Summing the normalized shares gives one. □

The optional adult distribution satisfies $0 \leq V_t \leq \overline{V}_t$. Unallocated capacity is not automatically spent.

## 12 Annual budget closure and fund accounting

Actual program distribution is

$$D_t = U_t + G_t^{\min} + H_t + G_t^{+} + T_t + V_t, \tag{26}$$

where $G_t^{+} = \overline{G}_t^{+}$ when $B_t > 0$ and 0 otherwise.

Current distributable resources, including a backstop if activated, are

$$C_t^Y = (1 - \phi_t)Y_t + Z_t. \tag{27}$$

Actual source amounts are defined after actual spending:

$$D_t^Y = \min\{D_t, C_t^Y\}, \tag{28}$$

$$D_t^W = D_t - D_t^Y, \tag{29}$$

$$E_t = C_t^Y - D_t^Y. \tag{30}$$

Here $E_t \geq 0$ is unused current revenue, which is transferred to the fund rather than silently disappearing from the accounts. The fund evolves as

$$W_{t+1} = W_t(1 + r_t) + \phi_t Y_t + E_t - D_t^W. \tag{31}$$

**Proposition 12.1** (Budget closure)**.** *Under the allocation rules above,*

$$0 \leq D_t \leq C_t, \qquad 0 \leq D_t^W \leq \overline{d}_t. \tag{32}$$

*Moreover, the accounting identity*

$$W_{t+1} = W_t(1 + r_t) + Y_t + Z_t - D_t \tag{33}$$

*holds.*

*Proof.* Mandatory spending satisfies $M_t \leq C_t$. Household spending satisfies $H_t \leq \beta_H(C_t - M_t)$. The three residual pools sum to at most $Q_t$, so total spending cannot exceed $C_t$. Since $C_t = C_t^Y + \overline{d}_t$, the residual definition $D_t^W = \left[D_t - C_t^Y\right]_+$ implies $D_t^W \leq \overline{d}_t$. Finally, $E_t = C_t^Y - D_t^Y$ and $D_t = D_t^Y + D_t^W$. Substitution into Equation (31) gives

$$W_{t+1} = W_t(1 + r_t) + \phi_t Y_t + C_t^Y - D_t = W_t(1 + r_t) + Y_t + Z_t - D_t.$$

□

**Corollary 12.2** (Positive and family-independent base)**.** *For every $c \in \mathcal{B}_t$,*

$$g_{c,t}^{(0)} \geq \underline{g}_t > 0, \tag{34}$$

*and this lower bound is independent of $K_{f,t}$, parental income, and education-module completion.*

## 13 Child-account dynamics

An SSCF account is the legal property of the child, administered in trust until staged access. It is not an asset of the parent, household, or general treasury. If accounts are invested inside the social fund, the public balance sheet must record an equal liability; otherwise the same assets would be counted both as public fund wealth and as private child wealth.

Let $A_{c,s}$ be the account balance at the start of date $s$, $q_{c,s} > -1$ the net credited return, $\tau_{c,s} \geq 0$ the top-up, and $x_{c,s} \geq 0$ the total amount paid out at the end of the period. The law of motion is

$$A_{c,s+1} = A_{c,s}(1 + q_{c,s}) + \tau_{c,s} - x_{c,s}, \tag{35}$$

subject to

$$0 \leq x_{c,s} \leq A_{c,s}(1 + q_{c,s}) + \tau_{c,s}. \tag{36}$$

This equation records withdrawals explicitly. Without $x_{c,s}$, a staged-release rule would repeatedly distribute claims against capital that remained unchanged on paper.

The public fund return $r_t$ and child-account return $q_{c,s}$ are distinct accounting objects. If accounts are externally custodied, $q_{c,s}$ is simply their net portfolio return. If internally custodied, the fund must segregate corresponding assets and liabilities.

## 14 Staged access without forfeiture

Let release ages satisfy $a_1 < a_2 < a_3$, for example 18, 21, and 25. Choose target shares $\eta_1, \eta_2, \eta_3 \geq 0$ with $\eta_1 + \eta_2 + \eta_3 = 1$. To avoid applying fractions repeatedly to an already reduced balance, define conditional release rates

$$k_j = \begin{cases} \dfrac{\eta_j}{1 - \sum_{m<j} \eta_m}, & j = 1, 2, \\ 1, & j = 3. \end{cases} \tag{37}$$

Let $P^-_{c,j}$ be the protected balance immediately before release $j$. The amount that vests out of protection is

$$R_{c,j} = k_j P^-_{c,j}, \qquad P^+_{c,j} = P^-_{c,j} - R_{c,j}. \tag{38}$$

Thus $k_1 = eta_1$, $k_2 = eta_2/(1 - \eta_1)$, and the final release transfers all remaining protected assets. Returns earned between milestones remain attached proportionally to the protected assets that produced them.

Financial education affects the form of early access, not ownership. Let

$$e_{c,j} = \sum_{m=1}^{J} \beta_m z_{c,m,j}, \qquad z_{c,m,j} \in \{0, 1\}, \quad \beta_m \geq 0, \quad \sum_m \beta_m = 1. \tag{39}$$

With guaranteed discretionary fraction $\delta \in (0, 1]$, the immediately unrestricted part of release $j < 3$ is

$$R^{\text{free}}_{c,j} = R_{c,j}[\delta + (1 - \delta)e_{c,j}]. \tag{40}$$

The remainder is placed in a restricted wallet usable for a broad statutory list that may include education, housing, mobility, tools, healthcare, debt avoidance, and enterprise. It is not forfeited. At $a_3$, all remaining restrictions expire regardless of module completion.

The education program must be free, accessible, multilingual, disability-accommodating, and subject to appeal. The modules should be short and practical. They cannot become a test of cultural conformity or an administrative device for permanent exclusion.

**Proposition 14.1** (No duplicate release)**.** *Ignoring payouts from already released wallets, the staged rule transfers the entire protected balance exactly once by age $a_3$.*

*Proof.* The first release leaves fraction $1 - \eta_1$. The second conditional rate transfers $\eta_2/(1 - \eta_1)$ of that protected balance, corresponding to the second target share after accounting for the first release. The third release rate equals one and therefore transfers the entire remaining protected balance. Each transfer is subtracted from $P^-_{c,j}$ before the next period. □

# 15 Real per-capita sustainability

Nominal fund non-decline is not an adequate definition of intergenerational sustainability. A fund can grow in currency units while shrinking after inflation or relative to the population it serves.

Define inflation and population growth by

$$1 + \pi_t = \frac{P_{t+1}}{P_t}, \qquad 1 + n_t = \frac{N_{t+1}}{N_t}, \tag{41}$$

where $\pi_t > -1$ and $n_t > -1$. Real per-capita fund wealth is

$$w_t = \frac{W_t}{P_t N_t}. \tag{42}$$

**Proposition 15.1** (Exact real per-capita condition)**.** *For $W_t > 0$, real per-capita fund wealth is non-decreasing, $w_{t+1} \geq w_t$, if and only if*

$$1 + r_t + \frac{\phi_t Y_t + E_t - D^W_t}{W_t} \geq (1 + \pi_t)(1 + n_t). \tag{43}$$

*Equivalently,*

$$r_t + \frac{\phi_t Y_t + E_t - D^W_t}{W_t} \geq \pi_t + n_t + \pi_t n_t. \tag{44}$$

*Proof.* By the fund law,

$$\frac{W_{t+1}}{W_t} = 1 + r_t + \frac{\phi_t Y_t + E_t - D^W_t}{W_t}.$$

Also,

$$\frac{P_{t+1} N_{t+1}}{P_t N_t} = (1 + \pi_t)(1 + n_t).$$

The condition $W_{t+1}/(P_{t+1}N_{t+1}) \geq W_t/(P_t N_t)$ is therefore equivalent to Equation (43). □

This is an ex-post accounting condition, not a promise that risky returns will realize. Ex-ante policy should use conservative expected real returns, rolling asset values, stress tests, and a stabilization reserve. A draw cap $\gamma_t$ is a prudential tool; it is not, by itself, proof of sustainability.

Where the relevant liability is more closely related to child cohorts than to total population, the same proposition may be stated using a transparent demographic index $L_t$ in place of $N_t$. The denominator should match the right being stabilized. Birth counts determine cohort bonuses directly; total population growth should not be used as an indirect substitute.

# Part III
# Calibration, Governance, and Evaluation

## 16 Illustrative calibration

The following values are arbitrary currency units (CU), not estimates for a particular country. They demonstrate accounting behavior.

Assume

$$Y_t = 100 \text{ billion}, \quad W_t = 500 \text{ billion}, \quad \phi_t = 0.40, \quad \gamma_t = 0.03, \quad \widetilde{W}_t = W_t,$$

and $r_t = 0.05$. Then

$$\overline{d}_t = 15 \text{ billion}, \qquad C_t^0 = 60 + 15 = 75 \text{ billion}.$$

Suppose

$$U_t = 25 \text{ billion}, \qquad B_t = 500{,}000, qquad \underline{g}_t = 10{,}000.$$

The aggregate statutory base is 5 billion and mandatory expenditure is

$$M_t = 25 + 5 = 30 \text{ billion}.$$

No backstop is required, so $Z_t = 0$ and $C_t = 75$ billion. Capacity after mandatory claims is 45 billion.

Let $\beta_H = 0.50$. The household-layer budget is 22.5 billion. If desired aggregate household support equals 20 billion, it is paid in full: $H_t = 20$ billion. Remaining capacity is

$$Q_t = 75 - 30 - 20 = 25 \text{ billion}.$$

Choose

$$\theta_G = 0.60, \qquad \theta_T = 0.20, \qquad \theta_V = 0.$$

The newborn bonus pool is 15 billion, so every newborn receives a bonus of

$$b_t^+ = \frac{15{,}000{,}000{,}000}{500{,}000} = 30{,}000.$$

The initial account is therefore 40,000 per newborn: a 10,000 protected base plus a 30,000 equal cohort bonus. The progressive top-up pool for all eligible child accounts is 5 billion. Total distribution is

$$D_t = 25 + 5 + 20 + 15 + 5 = 70 \text{ billion}.$$

Current distributable revenue is 60 billion, so

$$D_t^Y = 60 \text{ billion}, qquad D_t^W = 10 \text{ billion}.$$

The maximum draw is not taken merely because it is available. The closing fund is

$$W_{t+1} = 500(1.05) + 40 - 10 = 555 \text{ billion}.$$

### 16.1 Household-level illustration

Let $h_t = 5{,}000$, $s_t = 4{,}000$, $K^* = 2$, and $\lambda = 0.60$. Before any aggregate scaling,

$$\omega(1) = 1,$$
$$\omega(3) = 2.6,$$
$$\omega(6) = 3.3056.$$

Desired household supplements are therefore

$$\widehat{H}(1) = 9{,}000,$$
$$\widehat{H}(3) = 15{,}400,$$
$$\widehat{H}(6) = 18{,}222.40.$$

The taper bounds household support, while all newborns still receive the same cohort capital.

### 16.2 Account growth

With no later top-ups or withdrawals and a constant net return of 4 percent, a 40,000 initial account becomes

$$40{,}000(1.04)^{18} \approx 81{,}032.66$$

at age 18. This is a nominal illustration; actual policy must report inflation-adjusted values, fees, risk, and the distribution of realized returns.

## 17 Demographic and revenue stress tests

### 17.1 Birth shock

Suppose realized births rise from 500,000 to 650,000, while the child floor, capacity, statutory base, and desired aggregate household support remain as above. The base obligation rises to 6.5 billion, so

$$M_t = 31.5 \text{ billion}.$$

Capacity after mandatory claims is 43.5 billion. The household budget is 21.75 billion, still enough to pay the 20 billion desired amount. The new residual is 23.5 billion. Hence

$$\overline{G}_t^{+} = 0.60(23.5) = 14.1 \text{ billion},$$

and the per-newborn bonus becomes

$$b_t^{+} = \frac{14.1 \text{ billion}}{650{,}000} \approx 21{,}692.31.$$

Every child keeps the 10,000 base, and total newborn expenditure remains exactly within its authorized pool. The shock changes the bonus, not the legal minimum and not the budget identity.

### 17.2 Dedicated-revenue collapse

Suppose instead that $C_t^0 < M_t$. The backstop activates for exactly the shortfall $Z_t = M_t - C_t^0$. Household support and all residual pools fall to zero if no capacity remains. The system therefore protects essential childhood provision and the statutory capital base before optional layers. This ordering makes the political tradeoff visible: the state must either fund the promised right, amend it prospectively under valid legal rules, or admit default. It cannot conceal insolvency by assigning a zero residual to a nominally universal account.

# 18 Comparative statics and design choices

## 18.1 The statutory base $\underline{g}_t$

A higher base strengthens the independent child claim but raises mandatory expenditure linearly:

$$\frac{\partial G_t^{\min}}{\partial \underline{g}_t} = B_t.$$

It also increases the probability that the fiscal backstop is needed. The base should therefore be indexed and meaningful but calibrated to durable revenue, while variable prosperity is shared through the cohort bonus.

## 18.2 The household share $\beta_H$

A larger $\beta_H$ protects more residual capacity for household support. When desired support is below its cap, increasing $\beta_H$ has no effect. When the cap binds, it raises $H_t$ and correspondingly reduces the residual available for bonuses and top-ups.

## 18.3 The taper parameters $K^*$ and $\lambda$

Increasing $K^*$ gives full marginal household weight to more children. Increasing $\lambda$ slows the post-threshold taper. Neither parameter changes the child's statutory or cohort capital grant.

## 18.4 Saving and draw parameters

A larger $\phi_t$ reduces current-flow capacity and raises fund saving. A larger $\gamma_t$ increases maximum current capacity but does not require a larger draw. Actual $D_t^W$ is determined by actual expenditure after current resources are used. Both parameters should be evaluated against the real per-capita condition rather than a nominal balance target.

## 18.5 Progressive weights

Increasing $\kappa_y$ or $\kappa_d$ reallocates a fixed top-up pool; it does not increase aggregate spending. This is an advantage of normalization. Distributional choices are not allowed to masquerade as financing assumptions.

# 19 Governance and legal form

The mathematics defines claims and constraints. It cannot supply trustworthy institutions by itself.

## 19.1 Fund governance

The social wealth fund should have:

- statutory independence and a fiduciary mandate;
- professional management separated from benefit adjudication;
- public reporting of fees, risk, benchmarks, and voting policy;
- independent audit and legislative oversight;

- conflict-of-interest, procurement, and anti-corruption rules;
- a published asset-liability statement distinguishing public assets from child- account liabilities; and
- an emergency rule specifying when, by whom, and for how long ordinary draw limits may be altered.

Ring-fencing must be legal rather than rhetorical. Political raids can occur through direct withdrawal, mandated below-market investments, optimistic return assumptions, or the transfer of unrelated liabilities onto the fund. Governance rules must address all four.

### 19.2 Protection of child ownership

Accounts should be non-seizable by ordinary household creditors and unavailable as collateral before release. Guardians should not be able to redirect the balance. The law must specify beneficiary status in adoption, foster care, guardianship changes, death before release, and family dissolution. Administrative appeals should be simple, free, and independent.

### 19.3 Eligibility and migration

Eligibility choices are unavoidable. Citizenship-only rules can exclude long-term resident children; immediate eligibility can create political objections concerning short-term residence. The framework does not dictate a single rule, but it requires the rule to be prospective, administrable, and consistent with equal treatment. Possible designs include vesting by years of childhood residence, partial portability, and reciprocal arrangements across jurisdictions.

### 19.4 Privacy and data minimization

The universal base requires little means testing. Progressive top-ups require more data and therefore create privacy, stigma, and error costs. The state should use the smallest resource measure necessary, publish the formula, limit retention, prohibit unrelated use, and allow correction. Complexity should be justified by the additional equity it produces.

## 20 Macroeconomic and political-economy considerations

A national program of meaningful scale can affect asset prices, labor supply, savings, fertility, educational choices, migration, and credit markets. A grant usable for housing may partly capitalize into land prices if housing supply is constrained. A grant usable for tuition may be captured by providers if price regulation and supply do not respond. Capital access can increase entrepreneurship, but failure risk and market power shape who retains the gains.

The funding instruments also have different incidence and volatility. Resource rents are geographically concentrated and cyclical. Carbon charges should decline if climate policy succeeds. Land-value taxation is comparatively difficult to avoid but requires competent valuation. Inheritance taxation raises questions of valuation, avoidance, liquidity, family businesses, and cross-border mobility. The OECD concludes that well- designed inheritance taxation can raise revenue and advance equality of opportunity, while also emphasizing design tradeoffs and behavioral responses [10].

These differences argue against treating $Y_t$ as a single permanently stable stream. Empirical implementation should model each component separately, estimate covariance with fund returns and benefit needs, and stress-test severe joint shocks. A stabilization reserve and conservative statutory base are preferable to optimistic promises.

Political economy matters as much as portfolio theory. Universal components can build broad support, while progressive components direct more resources toward constrained children. Universality without progressivity may leave inherited hierarchy largely intact; progressivity without a common claim may become stigmatized and politically fragile. The SSCF deliberately combines the two while accounting for them separately.

# 21 Objections and replies

## 21.1 The framework penalizes saving or family provision

The equal base is not withdrawn from children whose parents save. Progressive top-ups may decline with household resources, but they are allocated from a distinct pool using a gradual bounded score. This preserves a universal ownership relation while directing additional help where capital constraints are greater.

## 21.2 Parents, not the state, should provide for children

Family duties and public duties are not mutually exclusive. Families already act inside legal, monetary, educational, and infrastructural systems. The proposal does not eliminate parental provision; it establishes a minimum social inheritance that cannot be erased by parental poverty, neglect, death, or family size.

## 21.3 Young adults will waste the grant

Some recipients will make poor decisions, as do some private heirs. That fact does not justify denying ownership only to people whose inheritance is publicly constituted. Staged access, accessible education, anti-fraud enforcement, and broad restricted uses can reduce predictable harms. Permanent paternalism would contradict the agency the capital is meant to support.

## 21.4 Public services are better than cash or capital

Often they are better for particular purposes. That is why the development floor has first priority. But services do not fully provide mobility, bargaining power, or the ability to initiate a project. The correct comparison is not services *or* capital; it is a carefully ordered bundle.

## 21.5 The common-inheritance category is politically manipulable

It is. The response is a statutory taxonomy, non-overlap rules, transparent incidence analysis, and democratic revision—not the fiction that existing property categories are pre-political. The framework intentionally limits its dedicated base rather than declaring all wealth socially claimable.

## 21.6 A universal program cannot repair historical injustice

Correct. It is not designed to settle all rectificatory claims. Universal common inheritance, progressive capability policy, and specific rectification should be debated on their own grounds and may operate simultaneously.

## 22 Empirical evaluation agenda

Implementation should begin with pilots or phased cohorts accompanied by pre-registered evaluation. Relevant outcomes include:

- childhood health, educational continuity, and housing stability;
- account accumulation, fees, returns, and loss dispersion;
- post-release education, mobility, home formation, enterprise, and debt;
- labor supply and job-search duration;
- fertility and household-formation responses;
- price capitalization in housing, tuition, and other eligible uses;
- take-up, administrative error, appeal outcomes, and distribution by protected characteristics;
- fund performance and real per-capita sustainability; and
- political durability and public understanding.

Evaluation should compare the full bundle with plausible alternatives: equivalent spending on services, unrestricted transfers, debt reduction, or tax credits. The question is not whether capital has benefits in isolation, but whether this architecture outperforms realistic uses of the same fiscal resources.

## 23 Limitations

This paper supplies a normative architecture and an internally balanced accounting model. It does not estimate behavioral elasticities, an optimal grant, or an optimal portfolio. It does not resolve the appropriate jurisdiction, the incidence of every funding instrument, or the interaction with existing welfare systems. It does not show that the SSCF is superior in every country, particularly where basic state capacity or essential services are absent.

The model also abstracts from uncertainty at the individual level. Returns differ, households move, administrative data contain errors, and some children require costly support not captured by a scalar adjustment. Aggregate budget closure is necessary but not sufficient for justice.

Finally, capital is not power-neutral. Asset managers, approved-use providers, lenders, and platform firms may attempt to capture the new balances. Consumer protection, competition policy, housing supply, and public provision are complements to the SSCF, not external details.

## 24 Conclusion

No generation creates its world from nothing. It inherits natural assets, accumulated knowledge, public infrastructure, legal institutions, ecological liabilities, and a distribution of property shaped by earlier decisions. Private effort and voluntary exchange remain morally and economically important, but they do not exhaust the sources of social wealth.

The Social Starting Capital Framework gives institutional form to that observation. It does not promise equal outcomes. It guarantees a child-development floor and a positive capital base; recognizes households without allowing family size to diminish the child's independent claim; distributes

bonuses and top-ups through exact pools; records withdrawals and liabilities honestly; and judges fund sustainability in real per-capita terms.

The proposal's moral test is simple: whether a child is treated as a future agent rather than merely as a dependent member of a fortunate or unfortunate household. Its fiscal test is equally simple: whether every promised unit is funded, every risk bearer is named, and future cohorts are included in the accounting. A society that claims human worth is not determined by birth should be willing to build institutions in which that claim has material content.

# A Implementation sequence

A cautious implementation can proceed in the following order:

1. Define the universal development package and measure its incremental cost.
2. Legislate the statutory base grant, index, eligibility rule, and fiscal backstop.
3. Establish the fund's asset-liability boundary and independent governance.
4. Dedicate non-overlapping revenue streams and publish incidence estimates.
5. Build account custody, registration, guardianship, and appeal systems.
6. Introduce household supplements under an aggregate cap.
7. Add normalized progressive top-ups only when data quality and privacy safeguards are adequate.
8. Pilot staged access and education rules, with all restrictions expiring at the final release age.
9. Publish annual budget closure, stress tests, and the real per-capita metric.
10. Evaluate outcomes against alternative uses of equal fiscal cost before scaling.

# B Compact algorithm

For each fiscal year $t$:

1. Observe $W_t$, $\widetilde{W}_t$, realized $Y_t$, registered $\mathcal{C}_t$, and finalized $\mathcal{B}_t$.
2. Compute $\overline{d}_t$, $C_t^0$, $U_t$, $G_t^{\min}$, $M_t$, $Z_t$, and $C_t$.
3. Compute desired household support, its aggregate cap, $\xi_t$, and actual $H_t$.
4. Compute $Q_t$ and the three residual pools.
5. Divide the newborn bonus by actual $B_t$; normalize top-up weights over actual eligible children.
6. Calculate actual $D_t$, then $D_t^Y$, $D_t^W$, and $E_t$.
7. Update the fund using Equation (31) and publish both sides of the accounting identity.
8. Credit individual accounts, record later withdrawals using Equation (35), and report segregated assets and liabilities.
9. Evaluate Equation (43) ex post and publish forward stress tests before setting the next year's parameters.

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